



Analytical investigation of coherence time from solid-state HHG

Guoxiang Luo¹, Jun Wang^{2,a}, and Xi Zhao^{3,b} 

¹ School of Physics and Electronics, Qiannan Normal University for Nationalities, Duyun 558000, Guizhou, China

² Institute of Atomic and Molecular Physics, Jilin University, Changchun, China

³ School of Physics and Information Technology, Shaanxi Normal University, Xi'an 710062, Shaanxi, China

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Abstract. We present an analytical investigation of the momentum-dependent electronic dephasing time $T_2(\mathbf{k})$ in crystalline solids driven by strong infrared laser fields. Our analysis of $T_2(k)$ reconstructed from MoS₂ and ZnO experimental solid-state high-order harmonic generation (HHG) data reveals that strong fields renormalize the scattering matrix elements via dynamical band mixing and phonon-assisted channels in polar semiconductors. Specifically, we show that the strikingly different momentum dependence of $T_2(\mathbf{k})$ in MoS₂ and ZnO arises from a cooperative interplay of dimensionality, orbital texture, and many-body interactions. In the two-dimensional MoS₂, reduced phase-space scaling and strong orbital mixing amplify angular anisotropies in the scattering processes, producing a highly non-monotonic $T_2(k)$. In contrast, the three-dimensional phase space of ZnO, together with its smoother orbital character and weaker excitonic corrections, yields a much more uniform and nearly monotonic dephasing profile. These results establish dimensionality and orbital hybridization as key drivers of material-specific, k -resolved dephasing dynamics. Our analysis indicates that reconstructing the momentum-dependent dephasing time $T_2(k)$ from solid-state HHG provides potential access to the measurement of ultrafast scattering dynamics, reveals the underlying orbital and band-geometry structure, enables accurate HHG-based band inversion, and offers material-specific parameters essential for predictive simulations and petahertz optoelectronic applications.

1 Introduction

The decoherence time T_2 is an important concept in the theory of open quantum systems and plays a foundational role across multiple disciplines of modern physics, including solid-state physics, quantum optics, ultrafast science, and quantum information [1–3]. In essence, T_2 characterizes the lifetime of phase coherence between quantum states, quantifying how rapidly the off-diagonal elements of the reduced density matrix decay under the influence of environmental interactions [4–6]. This loss of coherence governs the transition from quantum to classical behavior and sets the intrinsic limit for the persistence of quantum superpositions in realistic materials and [7–9].

In condensed-matter systems, T_2 directly determines the coherence of interband polarizations, excitons, valley pseudospins, and qubits hosted in semiconductors, two-dimensional materials, and defect centers [10–16]. The value and momentum dependence of $T_2(\mathbf{k})$ influence spectroscopic linewidths, optical response functions, and the efficiency of coherent control schemes. As such, T_2 is a key parameter in the design of

ultrafast optoelectronic and valleytronic devices, where long-lived coherence is desired, as well as in strongly correlated or topological materials, where dephasing provides insights into many-body scattering processes [17,18]. In quantum information science, T_2 sets the fundamental operational timescale for quantum gates and error correction, and its optimization remains a major driving force in the development of solid-state quantum technologies [7–9].

In parallel, the advent of strong-field physics in solids has introduced T_2 into a new regime of research: ultrafast electron dynamics driven by intense femtosecond and attosecond laser pulses [19–25]. Since the first experimental observation of solid-state HHG [23], it has become clear that T_2 plays an essential role in shaping harmonic yields and spectral phase [26,27]. The success of semiconductor Bloch equation (SBE) simulations incorporating phenomenological or momentum-resolved decoherence times has demonstrated that T_2 is indispensable for reproducing experimental HHG spectra, often outperforming real-space single-electron TDSE models and fully *ab initio* many-body calculations, this point was extensively discussed in previous work [27–31]. These comparisons highlight the importance of dephasing in strong-field-driven interband polarization and electron–hole recollision, posi-

^a e-mail: wangjun86@jlu.edu.cn (corresponding author)

^b e-mail: zhaoxi719@snnu.edu.cn (corresponding author)

tioning T_2 as a physically significant and experimentally relevant quantity for describing non-equilibrium dynamics in crystalline solids [26, 27, 32].

Despite its broad applicability, the physical interpretation of T_2 in the context of solid-state HHG remains under active discussion. Several fundamental questions persist: Is T_2 a rigorously defined physical observable or a phenomenological correction encapsulating unresolved many-body processes [33]? Why does solid-state HHG require extremely short decoherence times—on the order of only a few femtoseconds—much shorter than those in conventional linear-response spectroscopy? And why do momentum-dependent dephasing times $T_2(\mathbf{k})$ consistently yield dramatically improved agreement with experiments [26, 27, 32]? Addressing these questions is crucial not only for validating the SBE framework, but also for establishing a microscopic understanding of ultrafast dephasing in materials subjected to strong optical fields.

Recent progress has begun to shed light on these issues. Our previous work provided initial insight into the extremely short T_2 values required in HHG simulations and demonstrated that experimental HHG spectra in ZnO and MoS₂ can be used to reconstruct their momentum-resolved dephasing functions $T_2(\mathbf{k})$ [27, 34]. However, two key gaps remain: A strict mathematical formulation of T_2 within the SBE formalism has yet to be established, and the physical origin of the distinct k -dependent dephasing trends observed in different materials remains unexplained. The present study addresses these open questions, which—to the best of our knowledge—have not been explicitly discussed in prior literature. By elucidating the microscopic mechanisms governing $T_2(\mathbf{k})$, our work aims to advance the theoretical foundation of strong-field solid-state physics and deepen the understanding of ultrafast decoherence processes in materials.

The paper is organized as follows. In Section 2, we derive the momentum-dependent form of $T_2(\mathbf{k})$. Section 3 compares $T_2(\mathbf{k})$ in ZnO and MoS₂ under relevant experimental conditions, highlighting material-specific differences. Finally, we summarize our main conclusions, approximations as well as limitations of our model.

Atomic units are used throughout unless otherwise specified.

2 k -dependent dephasing time T_2

We start from a crystalline electronic system coupled both to a classical light field and to scattering baths (phonons, impurities, or other electrons):

$$H(t) = H_0 + H_{\text{light}}(t) + H_{\text{scatt}} + H_{\text{bath}},$$

$$H_0 = \sum_{n\mathbf{k}} \varepsilon_{n\mathbf{k}} c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}}. \quad (1)$$

Tracing out the bath degrees of freedom under the Born–Markov approximation yields the Redfield-type master equation for the reduced density matrix $\rho_s(t)$,

$$\frac{d\rho_s}{dt} = -\frac{i}{\hbar}[H_s(t), \rho_s] - \frac{1}{\hbar^2} \int_0^\infty d\tau \text{Tr}_{\text{bath}}[H_{\text{scatt}}(t), [H_{\text{scatt}}(t-\tau), \rho_s(t) \otimes \rho_{\text{bath}}]]. \quad (2)$$

Projecting Eq. (2) onto the interband polarization $p_{cv}(\mathbf{k}, t) = \langle c\mathbf{k} | \rho_s(t) | v\mathbf{k} \rangle$ gives

$$\frac{d}{dt} p_{cv}(\mathbf{k}, t) = -\frac{i}{\hbar} \Delta\varepsilon_{cv}(\mathbf{k}, t) p_{cv}(\mathbf{k}, t) + \mathcal{S}_{cv}(\mathbf{k}, t) - \int_0^\infty d\tau K_{cv}(\mathbf{k}, \tau) p_{cv}(\mathbf{k}, t-\tau), \quad (3)$$

where $\Delta\varepsilon_{cv}$ is the instantaneous interband energy spacing and K_{cv} is the bath memory kernel determined by H_{scatt} and bath correlators. If $K_{cv}(\mathbf{k}, \tau)$ decays rapidly compared with the timescale of p_{cv} , the Markov approximation leads to a local form:

$$\frac{d}{dt} p_{cv}(\mathbf{k}, t) = -\frac{i}{\hbar} \Delta\varepsilon_{cv}(\mathbf{k}, t) p_{cv}(\mathbf{k}, t) + \mathcal{S}_{cv}(\mathbf{k}, t) - \Gamma_{cv}(\mathbf{k}, t) p_{cv}(\mathbf{k}, t), \quad (4)$$

defining the dephasing rate and time as

$$\Gamma_{cv}(\mathbf{k}, t) \equiv \int_0^\infty d\tau K_{cv}(\mathbf{k}, \tau), \quad T_2(\mathbf{k}, t) = \Gamma_{cv}^{-1}(\mathbf{k}, t). \quad (5)$$

The microscopic expression for $\Gamma_{cv}(\mathbf{k}, t)$ follows from the generalized Fermi's golden rule:

$$\Gamma_{cv}^\alpha(\mathbf{k}, t) = \frac{2\pi}{\hbar} \sum_{n', \nu} \int \frac{d^d q}{(2\pi)^d} |M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}; t)|^2 \mathcal{F}^\alpha(f, N; t) \delta(\varepsilon_{n\mathbf{k}}(t) - \varepsilon_{n'\mathbf{k}+\mathbf{q}}(t) \pm \hbar\omega_{\mathbf{q}\nu}), \quad (6)$$

where α labels scattering channels (e–ph, e–e, impurity).

The above analysis shows that T_2 is not a fixed phenomenological constant but a dynamic quantity reflecting the competition between laser-driven coherent evolution and field-dressed scattering.

Equation (6) formally includes several scattering mechanisms, including electron–phonon, electron–electron, and impurity scattering. Under the experimental conditions relevant to the ZnO and MoS₂ HHG measurements considered here, electron–electron and electron–phonon scattering are expected to be the dominant contributions, while impurity scattering is typically weaker in high-quality crystalline samples.

A qualitative estimate suggests that electron–electron scattering can become particularly important under

strong-field excitation, since the laser field creates non-equilibrium carrier populations and enhances Coulomb-mediated scattering processes. Electron–phonon scattering, on the other hand, is constrained by phonon frequencies and thermal populations. Therefore electron–electron interactions are likely to provide the dominant contribution to the ultrafast dephasing times (on the order of a few femtoseconds) inferred from HHG experiments, with electron–phonon scattering providing an additional but generally smaller contribution.

The derivation above relies on the Markov approximation, which assumes that the bath correlation time is much shorter than the characteristic evolution time of the interband polarization. In typical semiconductors, the correlation time associated with electron–phonon and electron–electron scattering processes is on the order of tens to hundreds of attoseconds, whereas the strong-field-driven evolution of the interband polarization occurs on a femtosecond time scale. Under these conditions, the memory kernel $K_{cv}(\mathbf{k}, \tau)$ decays rapidly, justifying the Markov approximation used in Eq. (4).

For extremely strong fields or for ultrashort pulses approaching the attosecond regime, however, memory effects may become more relevant. In such cases the scattering kernel acquires a nonlocal time dependence and the evolution equation for p_{cv} becomes explicitly non-Markovian. While such corrections may modify the detailed temporal decay of the coherence, the momentum dependence of the scattering rate is still primarily governed by the microscopic matrix elements and phase-space factors in Eq. (6). Therefore the qualitative k -dependent trends of $T_2(\mathbf{k})$ discussed in this work are expected to remain robust.

3 Comparison of $T_2(k)$: MoS₂ vs ZnO

In the preceding section, we have rigorously demonstrated from first principles that the scattering rate (and hence the corresponding dephasing time) in solids is intrinsically a momentum-dependent quantity. Although we have formally derived the expressions for laser-induced electron–electron and electron–phonon scattering rates, their strict numerical evaluation requires first-principles packages [28, 35] that explicitly account for electron dynamics in strong laser fields, which is computationally demanding.

In our previous work [27, 34], we proposed a more efficient approach: By employing a machine-learning-based reconstruction scheme, we directly retrieved the $T_2(\mathbf{k})$ distributions of bulk ZnO along the Γ – M direction and monolayer MoS₂ along the K – K' direction from experimental HHG spectra. This method involves far less computational cost and allows direct comparison with experimentally measurable quantities.

It should be noted that the reconstructed $T_2(\mathbf{k})$ should be interpreted as an effective cycle-averaged dephasing time. Under strong laser driving the microscopic scattering rates can in principle depend explicitly on time through field-induced modifications of

the band structure, carrier populations, and screening. However, the reconstruction procedure extracts a time-independent effective $T_2(\mathbf{k})$ that reproduces the experimentally observed HHG spectra within the semiconductor Bloch equation framework. Consequently, the reconstructed $T_2(\mathbf{k})$ implicitly incorporates the time dependence of microscopic scattering processes into an effective rate averaged over the laser cycle and the HHG emission window.

Interestingly, the reconstructed $T_2(\mathbf{k})$ profiles of ZnO and MoS₂ exhibit strikingly different behaviors: in ZnO, $T_2(\mathbf{k})$ is approximately 1 fs near the Γ point but decreases rapidly to about 0.1 fs with increasing $|k|$, whereas in MoS₂, the $T_2(\mathbf{k})$ variation is much more complex, as shown in Fig. 1. In our earlier study, this qualitative difference was not discussed in detail. Here, we apply our present model to analyze and interpret the physical origin of these contrasting trends. Details about the experimental conditions can be found in [27, 34] and the references therein.

In the following, we analyze the distinct behaviors of ZnO and MoS₂ from four complementary perspectives: (i) dimensionality and phase-space scaling, (ii) matrix elements and orbital texture, (iii) excitonic effects, and (iv) intervalley scattering and momentum transfer.

3.1 Dimensionality and phase-space scaling

From Eq. (6) the phase-space prefactor scales as q_*^{d-1} . In 3D ($d = 3$) the extra q_*^2 factor smooths angular anisotropies: Small variations in the number of available scattering directions have less dramatic effect on the total Γ than in 2D. Mathematically:

$$\begin{aligned} \left. \frac{\partial \Gamma}{\partial k} \right|_{d=3} &\sim \int d\Omega_2 q_*^2 \cdots \quad (\text{smooth}), & \left. \frac{\partial \Gamma}{\partial k} \right|_{d=2} \\ &\sim \int d\theta q_* \cdots \quad (\text{angle-sensitive}). \end{aligned}$$

Hence, 2D materials can show strong *non-monotonic* $\Gamma(k)$ (and therefore $T_2(k)$) when the angular availability of solutions $q_*(k, \theta)$ and/or the matrix elements vary rapidly with k .

The dimensional dependence of Equation (6) can be further understood by examining the structure of the phase-space integration. For a scattering process constrained by energy conservation, the integration over the transferred momentum $\mathbf{q}$ reduces to an integration over a constant-energy manifold in momentum space,

$$\Gamma(\mathbf{k}) \propto \int d\Omega_{d-1} q_*^{d-1} |M(\mathbf{k}, \mathbf{q})|^2, \quad (7)$$

where $d\Omega_{d-1}$ denotes the angular measure in d dimensions and q_* is the magnitude of the momentum transfer determined by the energy conservation condition.

This expression reveals two distinct sources of dimensional dependence. First, the phase-space measure scales as q_*^{d-1} , reflecting the number of available scattering channels on the constant-energy surface. Second,

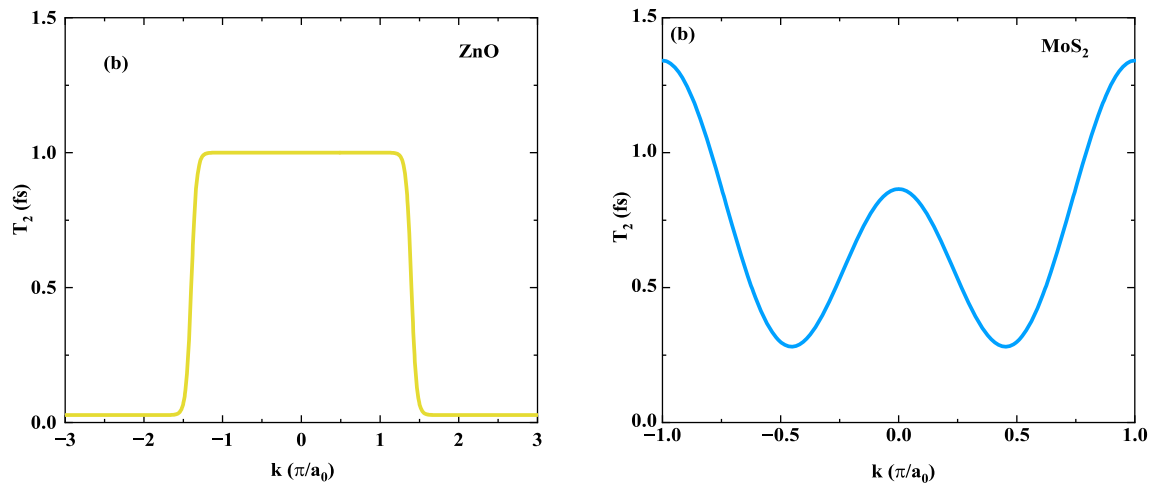


Fig. 1 This figure shows the reconstructed $T_2(k)$ from ZnO (Panle a) and MoS₂ (Panle b). Detail about the experimental schemes and laser parameters can be found in [26, 27, 34]

the angular integration averages over the anisotropy of the microscopic scattering matrix elements $M(\mathbf{k}, \mathbf{q})$.

In three dimensions, the larger phase-space factor (q_*^2) weights a broader range of scattering directions, which tends to average out angular variations of the matrix elements. Consequently, the resulting scattering rate $\Gamma(k)$ often exhibits a relatively smooth and monotonic dependence on the momentum magnitude.

In contrast, in two-dimensional systems the reduced phase-space measure (q_*) restricts the available scattering channels and preserves a stronger sensitivity to the angular structure of the matrix elements. As a result, anisotropic scattering processes can produce pronounced variations in $\Gamma(\mathbf{k})$, which may manifest as non-monotonic behavior of the effective dephasing time $T_2(\mathbf{k})$. This mechanism provides a natural explanation for the stronger momentum-dependent structures observed in reconstructed $T_2(k)$ for two-dimensional materials such as MoS₂, compared to the smoother behavior found in bulk materials like ZnO.

3.2 Matrix elements and orbital texture

The generic single-particle scattering matrix element introduced in Eq. (6) can be written in the Bloch basis as

$$M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}) = \langle u_{n\mathbf{k}} | \hat{O}^\alpha(\mathbf{q}) | u_{n'\mathbf{k}+\mathbf{q}} \rangle, \quad (8)$$

where $\hat{O}^\alpha(\mathbf{q})$ denotes the microscopic interaction vertex for channel α (e.g., $e^{-i\mathbf{q}\cdot\mathbf{r}}$ times a deformation/polar coupling for electron–phonon, or a screened Coulomb kernel for electron–electron scattering). The Bloch-periodic parts $u_{n\mathbf{k}}(\mathbf{r})$ carry the essential orbital information and depend on $\mathbf{k}$ through their expansion coefficients in a localized (atomic or Wannier) basis.

Expand $u_{n\mathbf{k}}$ in a localized orthonormal basis $\{\phi_\ell(\mathbf{r})\}$:

$$u_{n\mathbf{k}}(\mathbf{r}) = \sum_\ell c_{n\ell}(\mathbf{k}) \phi_\ell(\mathbf{r}).$$

Substituting into (8) gives

$$M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}) = \sum_{\ell, \ell'} c_{n\ell}^*(\mathbf{k}) c_{n'\ell'}(\mathbf{k} + \mathbf{q}) \langle \phi_\ell | \hat{O}^\alpha(\mathbf{q}) | \phi_{\ell'} \rangle. \quad (9)$$

Define the *vertex matrix* in the local basis

$$\mathcal{O}_{\ell\ell'}^\alpha(\mathbf{q}) \equiv \langle \phi_\ell | \hat{O}^\alpha(\mathbf{q}) | \phi_{\ell'} \rangle,$$

so that

$$M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}) = \sum_{\ell, \ell'} c_{n\ell}^*(\mathbf{k}) \mathcal{O}_{\ell\ell'}^\alpha(\mathbf{q}) c_{n'\ell'}(\mathbf{k} + \mathbf{q}). \quad (9')$$

If the localized basis is centered on lattice sites $\mathbf{R}_j$, each orbital carries a phase $\phi_{\ell j}(\mathbf{r}) = \phi_\ell(\mathbf{r} - \mathbf{R}_j)$. For a periodic lattice, the coefficients can be written including site phase factors; after summation one obtains a *structure factor* (or form factor) $F_{\ell\ell'}(\mathbf{k}, \mathbf{q})$ encoding interference between orbitals at different sites. Schematically,

$$M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}) = \sum_{\ell, \ell'} c_{n\ell}^*(\mathbf{k}) F_{\ell\ell'}(\mathbf{k}, \mathbf{q}) \tilde{\mathcal{O}}_{\ell\ell'}^\alpha(\mathbf{q}) c_{n'\ell'}(\mathbf{k} + \mathbf{q}),$$

where $F_{\ell\ell'}$ contains factors like $e^{i(\mathbf{k}+\mathbf{q})\cdot\mathbf{R}_{j'}-i\mathbf{k}\cdot\mathbf{R}_j}$ and $\tilde{\mathcal{O}}^\alpha$ denotes the intra-cell vertex. Rapid variation of the coefficients $c_{n\ell}(\mathbf{k})$ (or abrupt changes of orbital phase)

produces constructive or destructive interference in the sum, leading to strong k -dependence and possible nodes of M .

The scattering rate involves $|M|^2$. Using (9), one has

$$|M_{nn'}^\alpha(\mathbf{k}, \mathbf{q})|^2 = \sum_{\substack{\ell_1, \ell_2 \\ e\ell_1', \ell_2'}} c_{n\ell_1}^*(\mathbf{k}) c_{n\ell_2}(\mathbf{k}) \mathcal{O}_{\ell_1\ell_1'}^\alpha(\mathbf{q}) \mathcal{O}_{\ell_2\ell_2'}^{\alpha*}(\mathbf{q}) \\ c_{n'\ell_1'}^*(\mathbf{k} + \mathbf{q}) c_{n'\ell_2'}(\mathbf{k} + \mathbf{q}).$$

Thus the $\mathbf{k}$ -derivative of $|M|^2$ picks up terms proportional to derivatives of the coefficients $\partial_{\mathbf{k}} c_{n\ell}(\mathbf{k})$ and cross-terms between different orbitals. If the coefficients vary rapidly (large $|\partial_{\mathbf{k}} c_{n\ell}|$), the resulting $|M|^2$ can change rapidly with $\mathbf{k}$, producing peaks (enhanced scattering) or dips (suppressed scattering) in $\Gamma(\mathbf{k})$ after the $\mathbf{q}$ -integration in Eq. (6).

Consider a simple two-orbital model $\{\phi_A, \phi_B\}$ with

$$u_{n\mathbf{k}} = c_{nA}(\mathbf{k})\phi_A + c_{nB}(\mathbf{k})\phi_B.$$

If the relative phase/sign of c_{nA} and c_{nB} changes with $\mathbf{k}$, then overlaps such as $\langle u_{n\mathbf{k}} | u_{n'\mathbf{k}+\mathbf{q}} \rangle$ can vanish for some $\mathbf{k}$ (destructive interference). In that case $|M|^2$ has nodes and $\Gamma(\mathbf{k})$ is suppressed locally. Conversely, at $\mathbf{k}$ where coefficients align, $|M|^2$ is enhanced.

In monolayer MoS₂, the conduction and valence band Bloch functions near K and K' have strong and rapidly varying mixtures of Mo- d and S- p characters (This has been confirmed by many other works, such as Fig. 5 in [36]); numerically one finds $c_{n\ell}(\mathbf{k})$ that vary appreciably on scales of $|\Delta\mathbf{k}|$ comparable to the size of the valley. This rapid variation implies large $\partial_{\mathbf{k}} c_{n\ell}$ and nontrivial phase structure, hence strong $\mathbf{k}$ -dependence of $M(k, q)$ and possible non-monotonic behavior of $|M|^2$ as a function of k .

By contrast, in bulk ZnO the orbital weights change slowly across the Brillouin zone (Such results can be found in [37]): $c_{n\ell}(\mathbf{k})$ are smooth functions of $\mathbf{k}$, giving a matrix element that varies weakly with k . After integration over $\mathbf{q}$ the smoothing from the phase-space measure in 3D further reduces angular sensitivity (see dimensionality discussion).

The full scattering rate

$$\Gamma_n^\alpha(\mathbf{k}) \propto \int d^d q |M_{nn'}^\alpha(\mathbf{k}, \mathbf{q})|^2 \dots$$

inherits the strong k -dependence of $|M|^2$ when (i) orbital coefficients vary rapidly with k , (ii) the vertex $\mathcal{O}_{\ell\ell'}^\alpha(\mathbf{q})$ has nontrivial $\mathbf{q}$ -structure, and (iii) phase-space and energy conservation constraints select particular $\mathbf{q}$ -regions. Therefore materials with rapidly varying orbital texture (e.g., MoS₂) naturally exhibit strongly non-monotonic $\Gamma(\mathbf{k})$ and thus nontrivial $T_2(\mathbf{k})$, while materials with smooth orbital character (e.g., ZnO) show comparatively smooth $\Gamma(k)$.

To validate the analytical considerations presented above, we computed the dimensionally-integrated

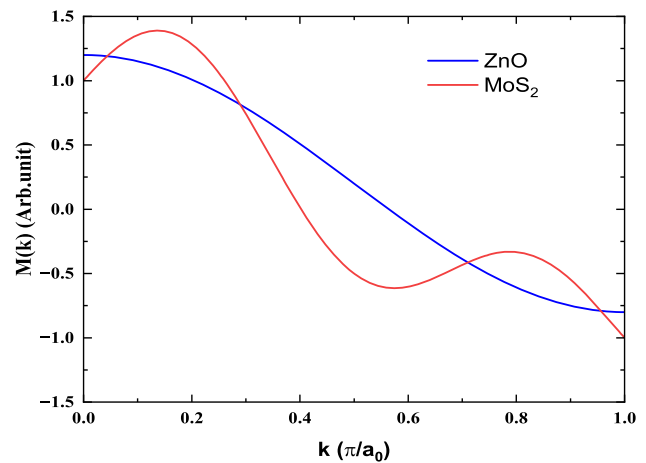


Fig. 2 This figure shows $M(k)$ of ZnO and MoS₂ from *ab initio* calculation

matrix elements $M(k)$ for ZnO and MoS₂ using the density-functional theory package OCTOPUS [38, 39]. For each system, we sampled 12 k -points along the relevant high-symmetry direction, using lattice parameters consistent with our previous work. As shown in Fig. 2, the resulting $M(k)$ for MoS₂ exhibits significantly stronger oscillatory behavior than that of ZnO, in agreement with our analytical predictions. This enhanced oscillation confirms the validity of our theoretical analysis.

3.3 Excitonic effects

Monolayer MoS₂ supports strongly bound electron-hole pairs (excitons). Microscopically, excitonic effects enter the two-particle (electron-hole) response function (polarization) which can be obtained from the Bethe-Salpeter equation (BSE). Denote the reducible electron-hole polarization by $P(\mathbf{q}, \omega)$ (frequency-domain retarded response). Formally, the BSE yields a spectral (pole) decomposition of the form

$$P(\mathbf{q}, \omega) = \sum_S \frac{|A_S(\mathbf{q})|^2}{\omega - \Omega_S(\mathbf{q}) + i\gamma_S(\mathbf{q})} \quad (10)$$

where S labels exciton eigenstates, $\Omega_S(\mathbf{q})$ are exciton dispersion energies, $A_S(\mathbf{q})$ are optical (or more generally transition) amplitudes (the exciton oscillator strength / spectral weight at momentum $\mathbf{q}$), and $\gamma_S(\mathbf{q})$ denotes the phenomenological exciton linewidth (inverse lifetime). Equation (10) follows from diagonalizing the particle-hole Hamiltonian (BSE kernel) and writing the reducible polarization in the eigenbasis (standard BSE spectral representation).

To see how $P(\mathbf{q}, \omega)$ affects interband coherence and the dephasing rate, consider a perturbative expression for the coherence self-energy (or scattering kernel) that couples an interband coherence at crystal momentum $\mathbf{k}$ to the two-particle excitations of the system. Schematically, the imaginary part of the coherence self-

energy Σ_{coh} (which determines the dephasing rate Γ via $\Gamma \sim 2 \Im \Sigma_{\text{coh}}$) can be written in lowest nontrivial order as

$$\Im \Sigma_{\text{coh}}(\mathbf{k}, \omega) \propto \sum_{\mathbf{q}} |M_{\text{eh}}(\mathbf{k}, \mathbf{q})|^2 \Im P(\mathbf{q}, \omega), \quad (11)$$

where $M_{\text{eh}}(\mathbf{k}, \mathbf{q})$ is an effective coupling matrix element that projects the single-particle interband coherence onto the two-particle excitation with center-of-mass momentum $\mathbf{q}$ (it contains combinations of the single-particle matrix elements and orbital coefficients introduced above). Equation (11) is the two-particle analog of the usual Golden-rule form: the coherence decays into two-particle excitations with a rate set by the coupling squared times the two-particle spectral weight $\Im P$.

Using the pole expansion (10) we have

$$\Im P(\mathbf{q}, \omega) = \sum_S |A_S(\mathbf{q})|^2 \frac{\gamma_S(\mathbf{q})}{[\omega - \Omega_S(\mathbf{q})]^2 + [\gamma_S(\mathbf{q})]^2}. \quad (12)$$

Substituting (12) into (11) gives

$$\Im \Sigma_{\text{coh}}(\mathbf{k}, \omega) \propto \sum_{S, \mathbf{q}} |M_{\text{eh}}(\mathbf{k}, \mathbf{q})|^2 |A_S(\mathbf{q})|^2 \frac{\gamma_S(\mathbf{q})}{[\omega - \Omega_S(\mathbf{q})]^2 + \gamma_S(\mathbf{q})^2}. \quad (13)$$

The physical content of (13) is transparent:

- Each excitonic pole S contributes a Lorentzian weight in frequency, centered at $\Omega_S(\mathbf{q})$ and with half-width $\gamma_S(\mathbf{q})$. The prefactor involves the product of the coupling $|M_{\text{eh}}|^2$ and the exciton oscillator strength $|A_S|^2$.
- For a fixed probing frequency ω (for instance, the interband transition energy or the laser-dressed transition relevant for HHG), only excitons with $\Omega_S(\mathbf{q}) \approx \omega$ give significant contributions. Near such resonance the Lorentzian amplitude scales like $\gamma_S/(\Delta^2 + \gamma_S^2)$ with $\Delta = \omega - \Omega_S$.
- Importantly, at exact resonance ($\Delta = 0$) the contribution is proportional to γ_S^{-1} in the sense that the Lorentzian peak height scales as $1/\gamma_S$ while its integrated spectral weight scales as a constant $\propto |A_S|^2$. However, the *imaginary part* which enters the scattering rate carries a factor $\gamma_S/(\Delta^2 + \gamma_S^2)$ as in (12).

To understand excitonic *stabilization* of coherence, consider two limiting situations for a given $\mathbf{q}$ where an exciton has spectral weight:

(i) *Narrow exciton (γ_S small) and resonant*: If ω is near $\Omega_S(\mathbf{q})$ and γ_S is small, the Lorentzian in (13) is sharply peaked. The *peak height* scales like $1/\gamma_S$ but its *width* in ω is $\sim \gamma_S$. For a fixed finite-temperature or finite-drive spectrum that samples a finite frequency window, a very narrow exciton tends to concentrate

spectral weight into a narrow frequency interval. If the system's relevant transition frequency ω does not sit exactly at Ω_S (or if the experimental probe averages over a frequency range), the effective $\Im \Sigma_{\text{coh}}$ seen by that probe can be *reduced* compared to the case of broad, continuum-like two-particle excitations. In other words, a narrow excitonic pole removes broadband two-particle spectral weight that would otherwise provide phase space for incoherent decay.

(ii) *Localized-in- $\mathbf{q}$ exciton*: Bound excitons in 2D materials are typically *localized in momentum space* (their center-of-mass wavefunction is broad in real space but narrow in $\mathbf{q}$ -space). Thus $|A_S(\mathbf{q})|^2$ is significant only in a small $\mathbf{q}$ -region. In (13) this localization means that only a restricted set of $\mathbf{q}$ contributes strongly to the sum; if the effective coupling $|M_{\text{eh}}(\mathbf{k}, \mathbf{q})|^2$ or the phase-space Jacobian disfavors these $\mathbf{q}$ for a given $\mathbf{k}$, then the total $\Im \Sigma_{\text{coh}}$ is suppressed. This is precisely the “stabilization” mechanism: Excitonic spectral weight does not provide broad continuum channels for incoherent decay across all $\mathbf{q}$, so the net scattering rate Γ can be reduced in regions of $\mathbf{k}$ where exciton weight dominates.

Putting these observations together: Narrow excitonic poles with small $\gamma_S(\mathbf{q})$ and localized $A_S(\mathbf{q})$ reduce the available broadband two-particle spectral density that would otherwise contribute to phase-randomizing scattering of interband coherences. Quantitatively, from (13) one can isolate a representative contribution from a single exciton near resonance:

$$\Gamma_{\text{exc}}(\mathbf{k}) \propto \sum_{\mathbf{q}} |M_{\text{eh}}(\mathbf{k}, \mathbf{q})|^2 |A_S(\mathbf{q})|^2 \frac{\gamma_S(\mathbf{q})}{[\omega - \Omega_S(\mathbf{q})]^2 + \gamma_S(\mathbf{q})^2}.$$

If γ_S is sufficiently small and $A_S(\mathbf{q})$ is sharply peaked at $\mathbf{q}_0$, the excitonic contribution to Γ is nonzero only in a narrow $(\mathbf{k}, \omega)$ -window and does not provide a broad background of dephasing. Consequently the *total* scattering rate (including non-excitonic continuum channels) is reduced in the $\mathbf{k}$ -region where the exciton dominates, i.e., $T_2(\mathbf{k}) = 1/\Gamma(\mathbf{k})$ is enhanced.

Finally, excitonic poles also modify the *real* part of the self-energy (dispersion/Lamb shift) via the principal-value integral associated with the pole. This yields a renormalization of transition energies which can further stabilize coherent oscillations by shifting resonances away from regions with large continuum spectral weight.

Summary.

Narrow, $\mathbf{q}$ -localized excitonic poles (γ_S small, $A_S(\mathbf{q})$ peaked) concentrate two-particle spectral weight into narrow regions of $(\mathbf{q}, \omega)$ and thereby reduce the broadband imaginary part that drives incoherent decay; this yields enhanced T_2 in the $\mathbf{k}$ -regions where the exciton carries spectral weight. In monolayer MoS_2 this mechanism is prominent owing to large exciton binding energies and sharp excitonic resonances; in ZnO the exciton-induced stabilization is typically weaker, consis-

tent with the differing $T_2(\mathbf{k})$ behavior observed experimentally and in our reconstructions.

We emphasize that in the present work the excitonic contribution to $\Gamma(\mathbf{k})$ is not calculated explicitly from first principles. Instead, the analysis based on the Bethe–Salpeter pole structure provides a qualitative microscopic mechanism illustrating how strongly bound excitons can modify the two-particle spectral density and thereby influence the effective dephasing rate.

The strength of this effect depends on both the exciton linewidth γ_S and the exciton binding energy. Narrower excitonic resonances (smaller γ_S) concentrate spectral weight into a narrow frequency window, while larger binding energies shift the excitonic poles away from the continuum. Both effects can reduce the broadband spectral density responsible for incoherent decay of interband polarization, thereby enhancing the effective T_2 in certain regions of the Brillouin zone. A fully quantitative treatment would require explicit Bethe–Salpeter calculations of exciton spectra and their coupling to strong-field-driven carriers, which is beyond the scope of the present analytical study.

3.4 Inter-valley scattering and large momentum transfer

At valley centers (e.g., K in MoS_2) intervalley scattering requires large momentum transfer $q \sim K$. If the relevant matrix elements decay for large q (e.g., deformation potential with short-range cutoff or screened Coulomb suppressed at large q), then the effective $|M(k, q_*)|^2$ at those q_* is small, suppressing Γ and increasing T_2 at valley points. Between valley and Γ regions, there can exist k such that (i) small-to-moderate q_* satisfy energy conservation and (ii) matrix elements and orbital mixing are favorable; this produces a *local maximum* of Γ (local minimum of T_2). This mechanism explains a non-monotonic shape $T_2(|k|)$ with enhanced dephasing at intermediate $|k|$ and increased T_2 at the endpoints.

We start from the golden-rule form used in Eq. (6), where we denote the energy mismatch

$$\Delta_{nn'}(\mathbf{k}, \mathbf{q}) \equiv \varepsilon_{n\mathbf{k}} - \varepsilon_{n'\mathbf{k}+\mathbf{q}} \pm \hbar\omega_{\mathbf{q}\nu}.$$

All dependence on microscopic orbital texture enters through the scattering matrix element

$$M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}) = \sum_{\ell\ell'} c_{n\ell}^*(\mathbf{k}) \mathcal{O}_{\ell\ell'}^\alpha(\mathbf{q}) c_{n'\ell'}(\mathbf{k} + \mathbf{q}),$$

as in the main text.

Root approximation of the $\mathbf{q}$ -integral. Assuming the delta function enforces isolated solutions $\mathbf{q} = \mathbf{q}_m^*(\mathbf{k})$ of $\Delta_{nn'}(\mathbf{k}, \mathbf{q}) = 0$ (nondegenerate roots), we may evaluate the $\mathbf{q}$ -integral by the standard change of variables (delta-function identity). For each root $\mathbf{q}_m^*$, one obtains

the contribution

$$\int d^d q |M|^2 \delta(\Delta) \approx \sum_m \frac{|M(\mathbf{k}, \mathbf{q}_m^*)|^2}{|\nabla_{\mathbf{q}} \Delta(\mathbf{k}, \mathbf{q})|_{\mathbf{q}=\mathbf{q}_m^*}}. \quad (14)$$

Hence, the scattering rate is approximated by

$$\Gamma_n^\alpha(\mathbf{k}) \approx \frac{2\pi}{\hbar} \sum_{n', \nu} \sum_m \frac{|M_{nn'}^\alpha(\mathbf{k}, \mathbf{q}_m^*)|^2 \mathcal{F}^\alpha(f, N)|_{\mathbf{q}_m^*}}{(2\pi)^d |\nabla_{\mathbf{q}} \Delta(\mathbf{k}, \mathbf{q})|_{\mathbf{q}_m^*}}. \quad (15)$$

Equation (15) makes manifest that $\Gamma(\mathbf{k})$ is controlled by three ingredients: The matrix element magnitude $|M(\mathbf{k}, \mathbf{q}_m^*)|^2$ evaluated at energy-conserving transfers the phase-space Jacobian $|\nabla_{\mathbf{q}} \Delta|$ and the occupation/thermal factor $\mathcal{F}$.

Large- q suppression at valley centers. At a valley center (e.g., $\mathbf{k}$ near K in MoS_2) intervalley processes require $|\mathbf{q}| \sim |\mathbf{K}|$ which is large. Typical microscopic vertices decay with increasing $|\mathbf{q}|$: For example, a short-range deformation potential or a localized intraband overlap leads to a Gaussian cutoff

$$|\mathcal{O}_{\ell\ell'}^\alpha(\mathbf{q})|^2 \propto e^{-q^2 \xi^2},$$

or screened Coulomb interactions behave as

$$|\mathcal{O}^\alpha(\mathbf{q})|^2 \propto \frac{1}{(q + q_s)^2},$$

so that for large q one has $|M(\mathbf{k}, \mathbf{q})|^2 \downarrow 0$. Thus for roots $\mathbf{q}_m^*$ with $|\mathbf{q}_m^*| \sim K$ the numerator of (15) is small, suppressing $\Gamma(\mathbf{k})$. Consequently $T_2(\mathbf{k}) = 1/\Gamma(\mathbf{k})$ is enhanced at valley points.

Existence of intermediate k with enhanced Γ . For $\mathbf{k}$ between valley and Γ regions, energy conservation may instead select small or moderate momentum transfers $\mathbf{q}_m^*(\mathbf{k})$ for which the vertex is not suppressed. Moreover, the orbital coefficients $c_{n\ell}(\mathbf{k})$ and $c_{n'\ell'}(\mathbf{k} + \mathbf{q})$ can interfere constructively for certain $\mathbf{k}$ (e.g., aligned phases or matching orbital weights), producing a locally large $|M(\mathbf{k}, \mathbf{q}_m^*)|^2$. If at the same time the Jacobian $|\nabla_{\mathbf{q}} \Delta|$ is not anomalously large, the corresponding term in (15) becomes large, producing a *local maximum* of $\Gamma(\mathbf{k})$ (and hence a local minimum of $T_2(\mathbf{k})$).

Simple 1D illustrative estimate. To make the mechanism transparent, consider a 1D cut (along $\Gamma - M$) and a single dominant scattering channel:

$$\Gamma(k) \approx C \sum_m \frac{|M(k, q_m^*(k))|^2}{|\partial_q \Delta(k, q)|_{q=q_m^*}},$$

with $C = 2\pi/(\hbar 2\pi)$. If we model the q -decay of the vertex by an exponential $|M|^2 \sim M_0^2 e^{-2\alpha q}$, then for

roots with $q^*(k)$ large we get

$$\Gamma(k) \sim \frac{M_0^2 e^{-2\alpha q^*(k)}}{|\partial_q \Delta|} \ll 1,$$

whereas for k with $q^*(k)$ small we obtain comparatively large $\Gamma(k)$. A non-monotonic $q^*(k)$ (determined by band dispersions) therefore directly translates into a non-monotonic $\Gamma(k)$.

Role of orbital texture derivatives. Using the local basis expansion

$$M(k, q) = \sum_{\ell\ell'} c_{n\ell}^*(k) \mathcal{O}_{\ell\ell'}(q) c_{n'\ell'}(k + q),$$

the derivative with respect to k reads

$$\partial_k M = \sum_{\ell\ell'} \left[(\partial_k c_{n\ell}^*(k)) \mathcal{O}_{\ell\ell'}(q) c_{n'\ell'}(k + q) + c_{n\ell}^*(k) \mathcal{O}_{\ell\ell'}(q) (\partial_k c_{n'\ell'}(k + q)) \right].$$

Rapid variations or phase windings in $c_{n\ell}(k)$ (large $|\partial_k c|$) thus produce large $\partial_k M$ and can create zeros or peaks of $|M|^2$ as a function of k . When such structure coincides with energy-conserving $q^*(k)$, the effect is amplified in $\Gamma(k)$ via (15).

Combining the root approximation (15), the large- q suppression of microscopic vertices and the k -dependence of the Bloch coefficients $c_{n\ell}(k)$, we obtain the following transparent picture: (i) valley centers requiring large q are protected from strong dephasing if the vertex decays for large q , leading to enhanced T_2 at valleys; (ii) intermediate k may admit small- q energy-conserving channels with constructive orbital interference, producing a local maximum of $\Gamma(k)$ (local minimum of T_2). This provides a microscopic explanation for the non-monotonic $T_2(|k|)$ profiles observed in materials with strong orbital texture (e.g., MoS₂), while materials with smooth orbital weights (e.g., ZnO) exhibit comparatively smooth $\Gamma(k)$ and monotonous $T_2(k)$.

4 Conclusion

In this work, we present a systematic microscopic study of the momentum-dependent interband dephasing time $T_2(\mathbf{k})$ in crystalline solids. Starting from a Born–Markov quantum master equation and tracing out phonon, electron, and impurity baths, we derive a generalized Fermi’s golden-rule expression for the scattering rate, $\Gamma(\mathbf{k}) = 1/T_2(\mathbf{k})$, establishing that T_2 has a clear physical meaning as the coherence decay rate induced by well-defined scattering processes.

By comparing monolayer MoS₂ and bulk ZnO, we reveal how material-specific properties, such as orbital mixing, excitonic effects, and dimensionality, lead to markedly different $T_2(\mathbf{k})$ profiles across the Brillouin

zone. The distinct momentum dependence of $T_2(\mathbf{k})$ in MoS₂ and ZnO originates from their different dimensionalities, orbital textures, and many-body environments. In two-dimensional MoS₂, the reduced phase-space scaling ($\propto q_*$) enhances the sensitivity of scattering to angular anisotropy and orbital mixing, producing a strongly non-monotonic $T_2(k)$ along K – K' . Rapid variations of the Bloch coefficients and pronounced excitonic effects further modulate the effective scattering matrix elements, yielding regions of both enhanced and suppressed dephasing. By contrast, in three-dimensional ZnO the additional q_*^2 phase-space factor smooths out these anisotropies, and the slowly varying orbital character and weaker excitonic corrections lead to a nearly monotonic and smoother $T_2(k)$ along Γ – M . Overall, the combination of dimensional scaling, orbital hybridization, and intervalley constraints quantitatively explains why and how ZnO and MoS₂ exhibits highly k -dependent dephasing dynamics.

These findings have several implications: $T_2(\mathbf{k})$ can be directly computed or experimentally reconstructed; its momentum dependence must be taken into account when using solid-state HHG to probe material properties; and its microscopic origin clarifies why previous $\mathbf{k}$ -dependent treatments of T_2 produce results more consistent with experiments. Overall, our work provides a potential microscopic foundation for the momentum-dependent dephasing time in solids and its role in strong-field light–matter interactions.

Our analysis relies on several approximations: weak system–bath coupling (Born), short bath correlation times relative to the system evolution (Markov), and neglect of rapidly oscillating cross-terms (secular). Additionally, non-equilibrium or very low-temperature baths may introduce higher-order many-body effects that could modify T_2 .

It is worth noting that although our model can quantitatively capture the essential $T_2(\mathbf{k})$ characteristics of both ZnO and MoS₂, its general applicability to other solid systems still requires further validation through systematic comparison between theoretical predictions and experimental observations.

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Data Availability Statement The codes used to generate the data presented in our manuscript are all fully based on publicly available software packages. The manuscript has associated data in a data repository.

Declarations

Conflict of interest The authors declare no conflict of interest.

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